National University of Computer and Emerging Sciences

Information Security (CS3002)

Student Name ______ Roll No ______ Section ______ Student Signature ______

Date: November 06, 2024

Course Instructor(s)

Dr. Sufian, Dr. Aqsa, Dr. Nausheen, Dr. Farrukh, Mr. Usman, Ms. Hira, Ms. Yusra

Sessional-II Exam

Total Time: 1 Hours
Total Marks: 45
Total Questions: 03

Semester: FA-2024
Campus: Karachi
Dept: CS/SE/AI

Instructions:

- **All questions must be answered in answer script and according to the sequence given in the question paper.**
- Illustrate/Design/Architect means you have to answers with the help of Diagram/Figure. **Written answers will not be graded where illustrations are required.** Most of the questions here require Illustrations (figures/algorithm) and unnecessary text will not be graded.
- Return the question paper.

CLO # 1: Explain key concepts of information security such as design principles.

Q1a: Illustrate how Diffie-Hellman key exchange is vulnerable to active adversarial attacks. **[05 marks]**

Q1b: Extend the original Diffie-Hellman key exchange to support three parties (Alice, Bob and Carol) without compromising the security. Show the negotiation of the key using as few steps as possible. In the end all the parties should end up with the key $\mathbf{K_{ABC}}$. **[05 marks]**

Q1c: Alice wants to send a secure message to Bob using the RSA encryption algorithm. To set up their communication, they agree on the following values: Prime numbers: p = 5 and q = 11, Encryption exponent: e = 3, Message to encrypt: m = 9 **[05 marks]**

i. Calculate the public key (e,n) for Alice and Bob's communication.
ii. Determine the private key (d,n) using the provided values.
iii. Encrypt the message m to find the ciphertext C.
iv. Decrypt the ciphertext C to retrieve the original message m.

CLO # 3: Analyze and apply various security and risk management tools for achieving information security and privacy.

Q2a: Alice, a data analyst at Company A, and Bob, a project manager at Company B, need to exchange sensitive reports over the internet securely. Create a diagram illustrating the process of generating and verifying a public-key certificate for Alice and Bob. Include the roles of the Certificate Authority (CA) and the steps involved in obtaining and verifying the certificates. **[05 marks]**

Q2b: Illustrate how Bob can send an encrypted report to Alice using public-key cryptography. Describe the encryption and decryption processes involved. **[05 marks]**

Q2c: The following is an X.509 certificate. **[05 marks]**

i. Who issues the certificate?
ii. Who is the owner of the certificate?
iii. Who generated the signature on this certificate, and how can this signature be verified?
iv. The public key contained in this certificate is based on the RSA algorithm. Using the RSA algorithm, to encrypt a message M, we calculate M^e mod n. What is the value of e and n in this public key? If a number is too large, you only need to write down its first four bytes.

```
Certificate:
 Data:
   Version: 3 (0x2)
   Serial Number:
     3d:0e:98:b2:bf:af:fa:9e:99:91:05:64:69:6e:11:2a
 Signature Algorithm: sha256WithRSAEncryption
   Issuer: C=US, O=Symantec Corporation,
           OU=Symantec Trust Network,
           CN=Symantec Class 3 EV SSL CA - G3
   Validity
      Not Before: Aug 14 00:00:00 2017 GMT
      Not After : Sep 13 23:59:59 2018 GMT
   Subject: ... C=US/postalCode=22230, ST=Virginia,
         L=Arlington/street=4201 Wilson Blvd,
         O=National Science Foundation, OU=DIS,
         CN=www.nsf.gov
   Subject Public Key Info:
     Public Key Algorithm: rsaEncryption
       Public-Key: (2048 bit)
         Modulus:
            00:ca:fb:26:78:06:25:b1:9e:67:1d:69:0b:10:06:
            cf:25:b6:7d:de:8e:56:80:e1:1c:38:52:62:43:fd:
            ...
      Exponent: 65537 (0x10001)
 Signature Algorithm: sha256WithRSAEncryption
     4b:0d:62:11:b4:dc:78:09:12:c1:1b:24:ff:98:43:58:1c:54:
     0a:34:be:8f:3f:12:8f:17:4a:fe:5b:26:13:1a:5f:a7:87:ad:
     ...
     ba:2c:10:c7:bc:8b:2c:15:6e:0c:d2:d0:8b:74:52:c8:ed:05:
     0b:9b:62:41
```

CLO # 4: Identify appropriate techniques to tackle and solve problems of real life in the discipline of information security.

Q3a: Define Malware. List three common types of malwares. Also differentiate between a virus and a worm, and provide two examples of each. **[05 marks]**

Q3b: Why botnet needs to be resilient against detection? Briefly explain how Domain Flux and Fast Flux algorithms adds resiliency against bot detection? **[05 marks]**

Q3c: A healthcare organization uses a management system to store patient records, treatment plans, and billing information. Access to this sensitive data is critical and must be tightly controlled.

i. Explain how ACLs control access to sensitive healthcare data. Describe how to set up ACLs to restrict treatment plan access to doctors and authorized nurses, with examples.
ii. Define capabilities for data sharing. Explain how a doctor can use them to share patient records with specialists, including implementation and access duration considerations.

[05 marks]